

*Metrics in DL is same as Classification metrics i.e accuracy, precision, recall, f1score
But in deep learning we use one more metric loss function*

Loss function:

- *Binary cross entropy : Bi classification*
- *Sparse categorical cross entropy : Multi classification*

Cross entropy:

$$\text{Entropy} = -p_{\text{yes}} * \log p_{\text{yes}} - p_{\text{no}} * \log p_{\text{no}}$$

$$p_{\text{no}} = (1 - p_{\text{yes}})$$

$$\text{Cross Entropy} = -p_{\text{yes}} * \log p_{\text{yes}} - (1 - p_{\text{yes}}) * \log(1 - p_{\text{yes}})$$

- *Generally Output network provides some values (logits), = $W * X + b$*
- *that values pass through activation function (probabilities) = $\text{sigmoid}(W * X + b)$*
- *activation function converts those values into probability values*
- *which ever is the highest probability that becomes the output*

Assume that we want perform animal object detection

There are 3 classes are there

- *Dog*
- *Cat*
- *Rabbit*

We are passing Dog image to the Neural network

There are 3 combinations

Actually Dog , Network predicted as Dog

Actually Dog , Network predicted as Cat

Actually Dog , Network predicted as Rabbit

Ground Truth(Original)	Predicted	Predicted probabillite	Cross entropy (disane between GT and PM)	
Dog(1)	Dog	0.8	0.2	Yes
	Cat	0.1	0.9	
	Rabbit	0.1	0.9	

DOG

The lower the loss, the more accurate the model

$y = [0.4, 0.4, 0.2]$
 $t = [0, 1, 0]$

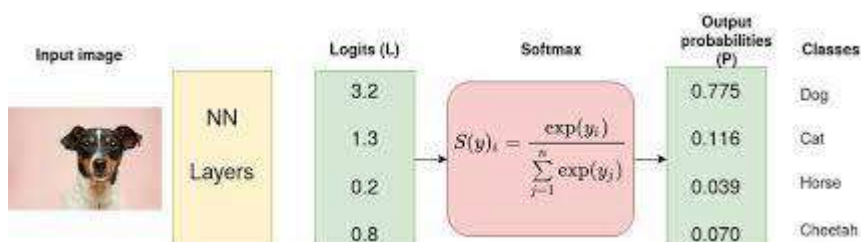
$L(y, t) = -0.4 \ln 0.4 - 1 \times \ln 0.4 - 0.2 \ln 0.2$
 $= 0.92$

HORSE

$y = [0.1, 0.2, 0.7]$
 $t = [0, 0, 1]$

$L(y, t) = -0.1 \ln 0.1 - 0.2 \ln 0.2 - 1 \times \ln 0.7$
 $= 0.36$

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- If Model will clearly Discriminate the lables with probabilties then Cross entrop low
- If Model will predict correct answers but the probabilties are not much differ then CE high
- If model predicts wrong answer then Cross entropy will be Huge
- Accuracy will tell the correct predictions with out related to model confidence
- Cross entropy will give the idea how much our model confident

Optimizer

Is used to speed up the process to find the weights

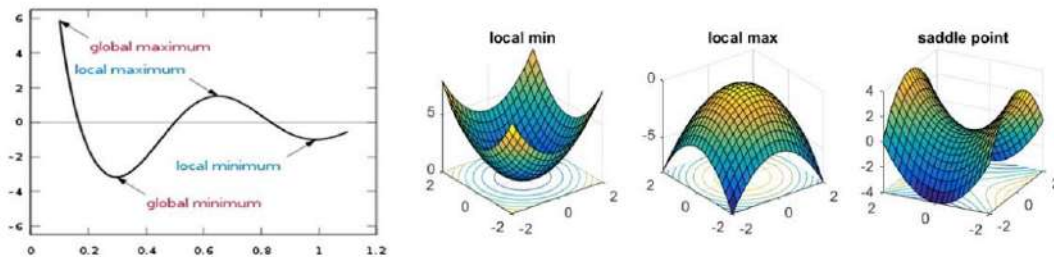
Generally to find the weights we have Grdient descent will use

But gradient descent fail when we have two minmum points

local minima and Global minima

Tutorial 13- Deep Learning

Global Minima vs Local Minima



Gradient descent

Gradient descent with Momentum : Global minima

Adaptive Gradient descent : Learning rate

Rms Prop : Decay rate (Momentum)

Adam (AdGrade + RmsProp)